Sunbeam Institute of Information Technology

Preparatory Assignments

C Programming Assignments

Q1. Write a program to input n numbers on command line argument and calculate maximum of them.

```java
class Assignment_1_p1{

    public static void main(String [] args ){

            if (args.length == 0) {
               System.out.println("Please provide numbers as command line arguments.");
                return;
            }

            int max = Integer.parseInt(args[0]);

            for (int i = 1; i < args.length; i++) {
                    if (Integer.parseInt(args[i]) > max) {
                        max = Integer.parseInt(args[i]);
                    }
            }

            System.out.println("Maximum number is: " + max);

    }

}
```

Q2. Write a program to calculate a Factorial of a number.

```java
import java.util.*;

class Assignment_1_p2{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int ans = factorial(n);
        System.out.println("Factorial Through recursion = " + ans);
        int fact = 1 ;
        for(int i=n; i>0 ; i--){
            fact*=i;
        }
        System.out.println("Factorial Through Looping = " + fact);

    }


    public static int factorial(int a ){
        if( a ==1 )
            return 1;

        return a*factorial(a-1);
    }
}
```

Q3. Write a program to calculate Fibonacci Series up to n numbers

```java
import java.util.*;

class Assignment_1_p3{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int prev =0 ;
        int curr = 1;
        System.out.println( "Febonacci series is as follows : \n"+prev+ "\n"+curr);
        for(int i = 0 ; i < n-2 ; i++){

            int temp = curr;
            curr = prev + curr ;
            prev = temp ;
            System.out.println(curr);
        }
    }

}
```

Q4. Write a program to calculate the grade of a student. There are five subjects. Marks in each subject are entered from keyboard. Assign grade based on the following rule:

Total Marks >= 90	Grade: Ex
90 > Total Marks >= 80	Grade: A
80 > Total Marks >= 70	Grade: B
70 > Total Marks >= 60	Grade: C
60 > Total Marks	Grade: F

```java
import java.util.*;

class  Assignment_1_p4{
    public static void main(String[] args) {
    if( args.length == 0 ){
        System.out.println("Marks Not Provided");
        return ;
    }

        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the Marks: ");
        int totalMarks = 0 ;
        for(int i = 0; i< 5 ; i++){
            //System.out.print("Enter the Marks of Subject "+i+" : ");

        int marks = Integer.parseInt(args[i]) ;//sc.nextInt();
            if(marks > 100){
                System.out.println("Enter appropriate Marks of Subject "+i+" Marks > 100 is not allowed ");
                i--;
                continue;
            }
            totalMarks += marks;
        }

        int avg = totalMarks / 5;

        boolean t = avg >= 90? true:false;

        if(t){
            System.out.print("Grade: Ex" );
        }
        else if(90>avg && avg >=80){
            System.out.print("Grade: A" );
        }
        else if(80>avg && avg >=70){
            System.out.print("Grade: B");
        }
        else if(70>avg && avg >=60){
            System.out.print("Grade: C" );
        }
        else if(60>avg){
            System.out.print("Grade: F" );
        }

    }

}
```

Q5. Write a program to check the input characters for uppercase, lowercase, number of digits and other characters. Display appropriate

message.

```java
import java.util.*;


class Assignment_1_p5{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int upper = 0, lower = 0, number = 0 ;

        System.out.print("Enter the String to be Evaluated : ");
        String str = sc.nextLine();

        for(char a : str.toCharArray()){

            if(Character.isLowerCase(a)){
                lower++;
            }
            else if (Character.isUpperCase(a)){
                upper++;
            }
            else if (Character.isDigit(a)){
                number++;
            }
        }

        System.out.println("The Upper case letter "+upper);

        System.out.println("The lower case letter "+lower);

        System.out.println("The digit letter "+number);

    }

}

```

Q6. Write a program to perform matrix multiplication.

```java
import java.util.*;
class Assignment_1_p6{

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        //For Matrix 1
        System.out.println(" Enter Size of the Matrix 1 ");
        System.out.print(" Enter row Size of the Matrix 1 ");
        int r1 =  sc.nextInt();
        System.out.print("\n Enter Column Size of the Matrix 1 ");
        int c1 = sc.nextInt();

        int [][] ma1 = new int[r1][c1];

        //For Matrix 2
        System.out.println("\n\n Enter Size of the Matrix 2 ");
        System.out.print(" Enter row Size of the Matrix 2 ");
        int r2 =  sc.nextInt();
        System.out.print("\n Enter Column Size of the Matrix 2 ");
        int c2 = sc.nextInt();

        if(c1!= r2 ){
            System.out.println("Does Not Follows the Matrix Multiplication Rule");
            return;
        }

        int [][] ma2 = new int[r2][c2];

        System.out.println("Enter the Elements of the matrix 1");
        for(int i = 0 ; i < r1 ;i++){
            for(int j = 0 ; j<c1 ; j++){
                System.out.print("\nEnter Element ma1: ");
                ma1[i][j]= sc.nextInt();
            }
        }

        System.out.println("Enter The Elements of the matrix 2");
        for(int i = 0 ; i < r2;i++){
            for(int j = 0 ; j<c2 ; j++){
                System.out.print("\nEnter Element ma2: ");
                ma2[i][j]= sc.nextInt();
            }
        }


        int [][] ans = new int[r1][c2];

        for(int i =0 ; i< r1 ; i++){

            for(int j =0 ; j< c2; j++){
                int element = 0 ;
                for(int k = 0 ; k < r2 ; k++){
                    element += ma1[i][k] * ma2[k][j] ;
                }
                ans[i][j]=element;
            }

        }

        System.out.println("The  Matrix1 is  : ");
        for(int i = 0 ; i < r1 ;i++){
            for(int j = 0 ; j<c1 ; j++){
                System.out.print(" "+ma1[i][j] +" ");

            }
            System.out.println();
        }

        System.out.println("The  Matrix2 is  : ");
        for(int i = 0 ; i < r2 ;i++){
            for(int j = 0 ; j<c2 ; j++){
                System.out.print(" "+ma2[i][j] +" ");

            }
            System.out.println();
        }


        System.out.println("The Final Matrix is  : ");
        for(int i = 0 ; i < r1 ;i++){
            for(int j = 0 ; j<c2 ; j++){
                System.out.print(" "+ans[i][j] +" ");

            }
            System.out.println();
        }


    }

}
```

Q7. Write a java program to accept a number from user using command line argument and display its table.

```java
class Assignment_1_p7{
    public static void main (String [] args ){
        if (args.length != 1) {
            System.out.println("Please provide a number as command line argument.");
            return;
        }

        int num = Integer.parseInt(args[0]);

        System.out.println("Table of " + num);
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " x " + i + " = " + (num * i));
        }
    }
}
```

Q8. Write a program to read the name of a student (studentName), roll Number (rollNo) and marks (totalMarks) obtained. rollNo may be an alphanumeric string. Display the data as read. Hint: Create a Student structure and write appropriate functions.

```java
import java.util.*;
class Assignment_1_p8{
    public static void main (String [] args ){
        Assignment_1_p8 s = new Assignment_1_p8();
        s.readData();
        s.displayData();

    }
    String studentName;
    String rollNo;
    double totalMarks;

    void readData() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        rollNo = sc.nextLine();

        System.out.print("Enter Total Marks: ");
        totalMarks = sc.nextDouble();
    }

    void displayData() {
        System.out.println("\nStudent Details:");
        System.out.println("Name: " + studentName);
        System.out.println("Roll No: " + rollNo);
        System.out.println("Total Marks: " + totalMarks);
    }

}
```

Q9. Accept an integer number and when the program is executed print the binary, octal and hexadecimal equivalent of the given number.
Sample Output: terminal> java Test Enter Number : 20
Given Number :20
Binary equivalent :10100
Octal equivalent :24
Hexadecimal equivalent :14
Hint: Use bitwise operators for binary conversion. Octal/Hexadecimal conversion to be done by repetitive division using recursion.

```java
import java.util.*;
class Assignment_1_p9{
    public static void main (String [] args ){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Number : ");
        int num = sc.nextInt();

        System.out.println("Given Number :" + num);

        System.out.print("Binary equivalent :");
        printBinary(num);

        System.out.print("\nOctal equivalent :");
        printOctal(num);

        System.out.print("\nHexadecimal equivalent :");
        printHex(num);

    }



    static void printBinary(int n) {
        if (n > 1)
            printBinary(n / 2);
        System.out.print(n % 2);
    }

    static void printOctal(int n) {
        if (n >= 8)
            printOctal(n / 8);
        System.out.print(n % 8);
    }

    static void printHex(int n) {
        if (n >= 16)
            printHex(n / 16);

        int rem = n % 16;
        if (rem < 10)
            System.out.print(rem);
        else
            System.out.print((char)(rem - 10 + 'A'));
    }
}
```

Q10. Read at most 10 names of students and store them into an array of char nameOfStudents[10][50]. Sort the array and display them back. Hint: Use qsort() method.

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return strcmp((char *)a, (char *)b);
}

int main() {
    char nameOfStudents[10][50];
    int n, i;

    printf("Enter number of students (max 10): ");
    scanf("%d", &n);

    printf("Enter names:\n");
    for(i = 0; i < n; i++) {
        scanf("%s", nameOfStudents[i]);
    }

    qsort(nameOfStudents, n, sizeof(nameOfStudents[0]), compare);

    printf("\nSorted Names:\n");
    for(i = 0; i < n; i++) {
        printf("%s\n", nameOfStudents[i]);
    }

    return 0;
}
```

Q11. Create a structure called Employee that includes three fields - a first name (type String), a last name (type String) and a monthly salary (double). Write functions to initialize the fields, print them and modify the values in the given object. Example methods:

- void emp_init(struct emp* e);
- void set_salary(struct emp *e, double sal);
- void emp_display(struct emp *e);

Write the test code in the main(). Create two emp objects and display each object's yearly salary. Then give each Employee a 10% raise and display each Employee's yearly salary again.

```c
#include <stdio.h>

struct emp {
    char firstName[50];
    char lastName[50];
    double salary;
};

void emp_init(struct emp *e, char *fn, char *ln, double sal) {
    strcpy(e->firstName, fn);
    strcpy(e->lastName, ln);
    e->salary = sal;
}

void set_salary(struct emp *e, double sal) {
    e->salary = sal;
}

void emp_display(struct emp *e) {
    printf("Name: %s %s\n", e->firstName, e->lastName);
    printf("Monthly Salary: %.2f\n", e->salary);
    printf("Yearly Salary: %.2f\n\n", e->salary * 12);
}

int main() {
    struct emp e1, e2;

    emp_init(&e1, "John", "Doe", 20000);
    emp_init(&e2, "Jane", "Smith", 25000);

    printf("Before Raise:\n");
    emp_display(&e1);
    emp_display(&e2);

    set_salary(&e1, e1.salary * 1.10);
    set_salary(&e2, e2.salary * 1.10);

    printf("After 10%% Raise:\n");
    emp_display(&e1);
    emp_display(&e2);

    return 0;
}
```

Q12. Write a Program to reverse the letters present in the given String. Do not use strrev() function.

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, len;

    printf("Enter string: ");
    gets(str);

    len = strlen(str);

    printf("Reversed String: ");
    for(i = len - 1; i >= 0; i--) {
        printf("%c", str[i]);
    }

    return 0;
}
```

Q13. Declare an Array of type char* and initialize it with a few strings (hardcoded). Display the strings which are duplicated in that array. (Hint: use strcmp())

```c
#include <stdio.h>
#include <string.h>

int main() {
    char *arr[] = {"apple", "banana", "orange", "apple", "banana", "grape"};
    int n = 6;

    printf("Duplicated Strings:\n");

    for(int i = 0; i < n; i++) {
        for(int j = i + 1; j < n; j++) {
            if(strcmp(arr[i], arr[j]) == 0) {
                printf("%s\n", arr[i]);
                break;
            }
        }
    }

    return 0;
}
```

Q14 Write a java code to check if string is palindrome.

```java
import java.util.*;
class Assignment_1_p14{
    public static void main (String [] args ){

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter string: ");
        String str = sc.nextLine();

        boolean isPalindrome = true;
        int len = str.length();

        for (int i = 0; i < len / 2; i++) {
            if (Character.toLowerCase(str.charAt(i)) !=
                Character.toLowerCase(str.charAt(len - 1 - i))) {
                isPalindrome = false;
                break;
            }
        }

        if (isPalindrome)
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

Q15. Input a string from the user. Count occurrences (case insensitive) of each alphabet in the string.

Sample output:

Input: Welcome to SunBeam.
Output:
A : 1
B : 1
C : 1
E : 3
L : 1
M : 2
N : 1
O : 2
S : 1
T : 1
U : 1
W : 1

```java
import java.util.*;
class Assignment_1_p15{
    public static void main (String [] args ){
        Scanner sc = new Scanner(System.in);
        System.out.print("Input: ");
        String str = sc.nextLine().toUpperCase();

        int[] count = new int[26];

        for(int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if(ch >= 'A' && ch <= 'Z') {
                count[ch - 'A']++;
            }
        }

        System.out.println("Output:");
        for(int i = 0; i < 26; i++) {
            if(count[i] > 0) {
                System.out.println((char)(i + 'A') + " : " + count[i]);
            }
        }

    }
}
```